Using Chemical Modeling to Understand the Impact of Temperature on Lithium-Ion Battery Performance

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Abstract

Lithium-ion batteries have been presented as one of the most efficient energy storage in the market. This type of battery is responsible for powering over 90% of the global grid market. [7] Based on the lithium-ion batteries' significant role in modern technology, understanding their efficiency during different temperature conditions is crucial to optimize their performance. Although there have been previous studies on the effects environmental factors have on lithium-ion batteries, our research utilizes PyBaMM and matplotlib, both software applications to run simulations that will measure the battery's performance based on its capacity and internal resistance. [6] The study aims to understand the connection between the battery's efficiency and temperature conditions. The final analysis concluded that the battery's capacity increased consistently alongside the rising temperatures, climbing to 280k. Afterwards reaching a plateau of roughly 0.68 Ah. While the battery's internal resistance steadily rises with the increasing temperature, peaking at 290k. After reaching 290K, the resistance drastically decreases, hitting its lowest point at 350k.

1 Introduction

With the demanding push to find alternatives to fossil fuels, lithium-ion batteries have emerged as a cornerstone technology powering many modern devices, from electric vehicles and mobile phones to laptops and renewable energy storage systems. Lithium-ion batteries are rechargeable, utilizing mechanisms to move lithium ions through the anode to the cathode allowing the battery during discharge and back when charging. [5] Although lithium-ion batteries are used in modern technology, this development is not a recent achievement. The beginning of lithium-ion battery technology started in the 1970s by M. Stanley Whittingham, but it took many prototypes and decades of research to refine the battery to its current state. After two decades of research, Sony and Asahi Kasei introduced rechargeable lithium-ion batteries as a commercial product. [4]

This breakthrough has revolutionized energy storage, helping present an alternative to fossil fuels and reducing greenhouse gas emissions. Their widespread reach in today's technology can be attributed to their high energy density, long lifespan, and lightweight nature, ideal for portable devices.

The rapid growth of lithium-ion batteries underscores the importance of optimizing their performance and understanding their limitations. This research paper aims to explore how environmental factors, particularly temperature, influence the performance of lithium-ion batteries. Previous studies have

established the impact that temperature has on battery efficiency, lifespan, and overall performance. Therefore, it is crucial to delve deeper into these aspects to enhance battery design.

2 Related Works

Previous studies used for this research include "A Review of Lithium-ion Battery Safety Concerns: The Issues, Strategies, and Testing Standards" which examines the effects of thermal temperature on lithium-ion batteries regarding their safety and performance [8]. The 2021 study explores lithium-ion batteries in the thermal temperature and our study goes into depth about how the battery's internal resistance and capacity react alongside ring temperatures. This is crucial for understanding how to create successfully safe and efficient batteries. Another 2018 study used as a reference is the "Predictive Modeling of Battery Degradation and Greenhouse Gas Emissions from U.S. State-level Electric Vehicle Operation" [3]. The study explains the process of developing predictive models for battery degradation and performance. These predictive models are significant in designing battery management systems to optimize performance and safety.

3 Methodology

To investigate the impact that varying temperatures have on lithium-ion batteries, the performance metrics measured must first be determined. For this paper, the two critical performance metrics will be capacity and internal resistance. These variables were decided due to their significant importance as indicators of the overall efficiency, reliability, and performance of lithium-ion batteries in various applications.

3.1 Performance Metrics

The first variable, capacity, refers to the amount of energy a battery can store and is measured in ampere-hours (Ah). Capacity is crucial when it comes to a battery's practical utility. A higher capacity showcases a longer duration of power supply, making this variable very important in determining battery performance.

The second variable, internal resistance, refers to the opposition of the flow of a current in the battery and is measured in ohms. Zero internal resistance is most desirable since there would be the most efficient energy transfer from the battery, minimizing any losses due to heat. However, since there is always an internal resistance, a good battery should always aim to have low levels of internal resistance. It should be noted that repeated use of a battery over a long period will cause the battery's internal components to degrade and wear "due to rust and corrosion inside the electrodes," thereby increasing the internal resistance of the battery and making it less efficient [1].

3.2 Environment of Simulation

Due to the challenges associated with physical investigation, the data for this study will be extracted from a computational method. The data for this research was generated using PyBaMM, a Python-based simulator specifically measuring lithium-ion battery performance under varying temperatures.

PyBaMM can create a virtual lithium-ion battery that mimics one in the real world; utilizing built-in methods, data regarding the two variables can be extracted without needing to physically measure them with a real lithium-ion battery. This data extraction will be repeated with varying temperatures.

3.3 Temperature Range and Parameters

The Python-based simulator specifically explores the temperature, measured in Kelvin, which will range from 250 to 350 in this study. Choosing this range was important to replicate the natural temperatures under which lithium-ion batteries perform in ordinary circumstances; simulating both realistic colder and hotter environments while extracting data from each of them will provide a good data set with information that could be applied to the real world.

	Temperature	Capacity	Internal Resistance
0	250	0.320094	0.737535
1	260	0.523247	0.807640
2	270	0.662300	0.867636
3	280	0.673295	0.917998
4	290	0.680616	0.956067
5	300	0.680616	0.848601
6	310	0.680616	0.779865
7	320	0.680616	0.730946
8	330	0.680616	0.694632
9	340	0.680616	0.668237
10	350	0.680616	0.649742

Table 1: Simulation results: effect of temperature on battery capacity and internal resistance

3.4 Data Analysis and Visualization of Data

All of the data will be put into a data frame, with 11 columns for each temperature tested, and 3 rows with each of the variables (temperature included).

Utilizing matplotlib, these results will be processed and visualized into two separate graphs, showing the data in a more digestible manner.

Each graph will have temperature in Kelvin as the independent variable on the x-axis. Along with capacity (Ah) and internal resistance (ohms) measured on the y-axis on their respective graphs.

These visuals can highlight the relationship between temperature and the variables in both a qualitative and quantitative manner. Acknowledging and understanding trends and patterns within the graphs will lead to potential conclusions regarding the effect of temperatures on holistic battery performance.

3.5 Data Validation and Interpretation

To ensure the reliability of the simulation results, the data was validated against known theoretical models and existing experimental findings in the literature. To achieve this comparison, we used data from Chandan Chetri and Sheldon Williamson's study on "Accelerated Battery Capacity Degradation using Voltage-sag Analysis under Sub-zero Fast Discharging Conditions" [2]. This step was crucial in verifying that the simulated trends accurately reflect the real-world behavior of lithium-ion batteries under different thermal conditions. The validated results were then interpreted in the context of practical applications, providing guidance for battery design and usage in environments where temperature is a critical factor.

4 Results

This section presents the results of the PyBaMM lithium-ion Single Particle Model (SPM) conducted to analyze the impact of temperature on the performance of lithium-ion batteries. The primary performance metrics considered in this study were battery capacity and internal resistance within the simulated range of 250K to 350K. The Mean Squared Error (MSE) was also calculated to assess the model's accuracy against the literature data.

The graph above (Figure 1) reveals a non-linear correlation between temperature and capacity. Initially, a pronounced increase in capacity is observed with rising temperatures, which suggests an improved battery performance at higher temperatures. However, beyond approximately 280K, the capacity curve flattens around 0.68 Ah. This suggests that there exists an optimal temperature range where the battery operates most efficiently in terms of capacity, and beyond which further increases yield diminishing returns.

The provided data in the figure above (Figure 2) illustrates the complex connection between temperature and internal resistance. A notable increase in internal resistance is observed with increasing temperature throughout the simulated range, until a peak of approximately 0.96 ohms at 290 K .

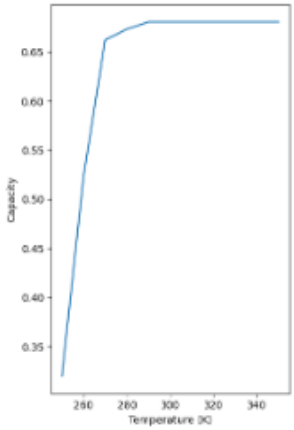


Figure 1: The variation in battery capacity across the simulated temperature range of 250K to 350K.

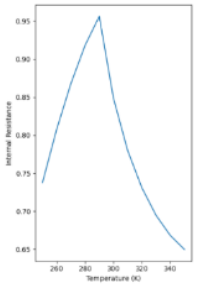


Figure 2: The variation in internal resistance across the simulated temperature range of 250K to 350K.

This suggests that higher temperatures lead to increased energy losses within the battery. Beyond that temperature, the internal resistance begins decreasing reaching a minimal value of around 0.65 ohms at 350K. This finding emphasizes the importance of effective thermal management systems to mitigate the detrimental effects of elevated temperatures on battery performance. By preventing excessive heating, it is possible to maintain lower internal resistance for safer use and better battery efficiency.

Based on the analysis of the simulated Lithium-ion battery's capacity and internal resistance across the temperature range of 250K to 350K, it is evident that an association of these factors influences the optimal battery performance. Our results highlight how higher temperatures generally lead to increased capacity, but the concurrent rise in internal resistance poses a challenge. The observed plateau in capacity and the decrease in internal resistance beyond 290K suggest a potentially favorable temperature range between 290K and 300K in which the battery performance is optimal (according to our simulation).

The figures (Table 2) and (Table 3) show the MSE for the capacity and internal resistance metrics across the specified temperature range.

5 Discussion

The study explored the dependent nature of the temperature of lithium-ion batteries using the Single Particle Model (SPM) in PyBaMM, highlighting two performance metrics: internal resistance and capacity. There is a temperature range of 250K to 350K. [8]

Table 2: Literature Actual Battery Capacity VS. Predicted Experimented Capacity

Temperature	Actual Capacity	Experimented Capacity	MSE
250	0.320094	0.4765	0.024463
260	0.523247	0.5012	0.000540
270	0.662300	0.7678	0.011130
280	0.673295	0.8515	0.031757
290	0.680616	0.8613	0.032719
300	0.680616	0.8920	0.044472
310	0.680616	0.9120	0.054936
320	0.680616	0.9130	0.054936
330	0.680616	1.1000	0.175883
340	0.680616	1.1500	0.220321
350	0.680616	1.1520	0.220321

Table 3: Literature Actual Internal Resistance VS. Predicted Experimented Internal Resistance

Temperature	Actual Internal Resistance	Experimented Internal Resistance	MSE
250	0.737535	1.0878	0.122686
260	0.807640	1.0987	0.084716
270	0.867636	1.1130	0.060203
280	0.917998	1.1240	0.042437
290	0.956067	1.2132	0.066117
300	0.848601	1.1987	0.122569
310	0.779865	1.1830	0.162518
320	0.730946	1.1656	0.188924
330	0.694632	1.1789	0.234515
340	0.668237	1.1870	0.269115
350	0.649742	1.1850	0.269115

5.1 Capacity Trends

The results of the battery's capacity (Figure 1) exhibited a non-linear relationship with temperature. As the temperature increased, capacity increased sharply with temperature, peaking around 280K before reaching a plateau at approximately 0.68 Ah. This trend aligns with existing literature that suggests battery capacity improves with temperature due to enhanced ionic mobility and reduced charge-transfer resistance at the electrode/electrolyte interface.

However, the capacity curve's flattening beyond 280K suggests diminishing returns with further temperature increases. This plateau may be attributed to the chemical processes within the battery, where additional thermal energy no longer translates to significant improvements in capacity. The identification of this plateau is crucial for battery operation, as it indicates an optimal temperature range for maximizing capacity without risking thermal degradation.

The MSE values, calculated between the simulated data and literature values, further validate the accuracy of the model. The low MSE values, particularly at higher temperatures, suggest that the model accurately captures the temperature-dependent capacity trends observed in practical scenarios. (Table 2)

5.2 Internal Resistance Trends

The results of the battery's internal resistance (Figure 2) indicate an abstract relationship between temperature and internal resistance. Initially, internal resistance increases with temperature, peaking at approximately 290K.

Beyond 290K, however, internal resistance begins to decrease, reaching its lowest value at 350K. This reduction in resistance at higher temperatures might be explained by the enhanced ionic conductivity within the electrolyte. Despite this decrease, it is crucial to note that operating the battery at excessively high temperatures can still be detrimental, as it may accelerate aging processes and reduce the battery's overall lifespan.

The MSE values calculated for internal resistance show a higher deviation compared to capacity, although still great for the model's accuracy, particularly at lower and higher temperatures. (Table 3)

6 Conclusion

The intricate relationship between temperature and lithium-ion battery performance is a critical factor influencing the overall efficiency and lifespan of these energy storage devices. This study demonstrated the connection between temperature, capacity, and internal resistance. Although higher temperatures can boost capacity—until plateauing at a relatively high level—, the accompanying non-linear behavior of internal resistance, initially increasing before decreasing, underscores the importance of precise thermal management.

Based on the simulated data, a potential optimal temperature range for maximizing capacity and minimizing internal resistance lies between 290K and 300K. Nevertheless, it is crucial to acknowledge that real-world conditions, including battery chemistry, cell design, and specific application requirements, can significantly influence optimal operating parameters. To gain a more comprehensive understanding, future research should investigate a wider range of battery chemistries, and consider additional performance metrics such as cycle life and safety, in parallel to both modeling and experimental data. Ultimately, such a holistic approach would be useful to develop advanced thermal management strategies, and thus optimize lithium-ion batteries' performance.

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